

Graph Neural Networks (GNN) - Notes

Graph Neural Networks (GNNs) are a class of deep learning models designed to work directly on graph-structured data.

They generalize neural networks to handle relationships and connections between data points (nodes).

--- Key Concepts ---

1. Graph Basics

- Nodes (vertices): represent entities.
- Edges: represent relationships between entities.
- Graph: combination of nodes and edges.

2. GNN Working

- Message Passing: Each node aggregates information from its neighbors.
- Update: Node updates its representation based on aggregated information.
- Readout: Graph-level representation is generated (for classification/regression).

3. Types of GNN Models

- GCN (Graph Convolutional Network): Extends convolution to graphs.
- GraphSAGE: Samples and aggregates neighbors instead of full graph processing.
- GAT (Graph Attention Network): Uses attention mechanism to weigh neighbors differently.
- GGNN (Gated Graph Neural Network): Uses gating mechanisms (GRUs) for updates.
- PinSage, ChebNet, etc. are further variations.

4. Applications

- Social networks (friend recommendation).
- Chemistry (molecule property prediction).
- Traffic prediction.
- Recommendation systems.
- Fraud detection.

5. General Steps to Build a GNN

- Define the graph structure (nodes, edges, features).
- Choose a GNN architecture (GCN, GraphSAGE, GAT, etc.).
- Train the model using supervised/unsupervised learning.
- Evaluate performance on classification/regression tasks.